

The FPGA extension

The FPGA extension allows users to export and test projects on FPGA boards without being familiar with the FPGA and its development tools. Projects can range from elementary to simple, but these must be meaningful embedded systems. This allows students to focus only on the essential steps of design. They can visualise their results immediately from the lab itself.

VHDL code generated by Deeds is observable, modifiable and reusable. It can thus act as the path and stimulus for transition from schematic entry to hardware description languages.

A microcomputer based digital signal generator

Let us discuss an interesting use case where the general-purpose 8-bit microcomputer is programmed to work as a DSP. Here, the software is used to implement a low-frequency digital waveform generator that fits on FPGA boards.

For this we need a sine-wave generator, a modulator and a display system. Output of the digital-signal generator could be used to drive a row of LEDs available on the board. The resulting display will resemble movements of a pendulum and would be quite interesting for students.

Output of this digital system can be observed on the timing diagram almost as it were an analogue signal.

are plotted by a pen at the empty square, forming a chart logging generated waveform. Input side of DAC acts as a parallel register and output side as a data-display device. Data gets loaded and carried to the data-display device at the positive clock edge.

DAC register is reset and output chart is cleared when clear input is activated. It should be noted that no analogue output is really generated.

Advanced ROM. Component library has been improved with two new ROM types. Contents at ROM could be edited, giving users the possibility of initialising the memory with a sampled

